RESEARCH AND TECHNICAL REPORT

# Comparative Landscape of AI Governance and Runtime Authorization

An Evidence-Bounded Review of Guardrails, Agent Security, Formal Verification, Governance Standards and the Organic Intelligence Protocol

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**Organic Intelligence Protocol**

OIP

**Prepared and Authored by**

## Sudharsan Das

Lead Architect, Organic Intelligence Protocol

Research Author


Responsible for the conceptual direction and research documentation of OIP presented in this report, including its methodological framing, evidence boundaries and research-stage status.


Document Reference: OIP-COMP-2026-V1

Research-stage comparative review

7 November 2026



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| | |
|---|---|
| **Document Title** | Comparative Landscape of AI Governance and Runtime Authorization |
| **Document Reference** | OIP-COMP-2026-V1 |
| **Project** | Organic Intelligence Protocol (OIP) |
| **Author** | Sudharsan Das |
| **Project Role** | Lead Architect, Organic Intelligence Protocol |
| **Document Type** | Research-stage comparative report |
| **Issue Date** | 7 November 2026 |
| **Evidence Cut-off** | 7 November 2026 |
| **Validation Status** | Comparative review; not independent validation |
| **Copyright Holder** | Sudharsan Das, for original material in this report |

### Document Purpose

This report provides an evidence-bounded comparison of selected AI governance, guardrail, runtime security, formal verification and regulatory approaches.

It also documents the current research-stage position of the Organic Intelligence Protocol.

The report is not a product certification, independent security audit, legal opinion or scientific validation.

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## 1 Executive Summary

AI governance and agent-security systems address different parts of the problem of controlling AI behaviour.

Some systems apply content policies and model safeguards. Others inspect tool calls, constrain execution environments, monitor workloads, verify mathematical properties or establish organizational governance requirements.

These approaches should not be treated as interchangeable.

This report compares selected systems and frameworks across six areas:

1. AI guardrails and model-level safeguards.
2. Runtime authorization and agent security.
3. Evidence handling, provenance and missingness.
4. Formal verification and mathematical assurance.
5. Governance standards and regulatory requirements.

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6. The documented research-stage position of OIP.

The comparison is based on identified primary-source documentation and the OIP research record available for this review. It is a representative landscape, not an exhaustive inventory of the global AI governance field.

The central finding is that these approaches operate at different intervention points. A content safeguard, a tool permission boundary, a neural-network verifier and an organizational management standard answer different questions.

OIP is evaluated within its documented scope. Its research-stage methodology addresses evidence qualification, measurement authorization, missingness controls and fail-closed decision rules.

Its wider scientific and operational validity remains unestablished. This report does not claim that OIP is technically superior, globally unique or a validated replacement for existing systems.

## 2 Author and Research Context

### 2.1 Author

**Sudharsan Das**

**Project Role:** Lead Architect, Organic Intelligence Protocol.

Sudharsan Das is the author of this report and the Lead Architect responsible for the OIP research direction and methodological documentation described here.

The report records his work in organizing the OIP research framework, documenting its conceptual architecture, maintaining evidence boundaries and identifying unresolved methodological questions.

### 2.2 Why Authorship Is Identified

Clear authorship is important for research provenance, document accountability and version control.

The author identification establishes:

- Who prepared and authored this comparative report.
- Who is identified as Lead Architect of OIP.
- Which project and research documentation are discussed.
- Which claims remain subject to independent review.

The author's role should not be confused with external scientific endorsement, institutional affiliation, independent validation or regulatory recognition.

## 3 Purpose and Scope

### 3.1 Purpose

The purpose of this report is to establish a structured comparison of selected AI governance, runtime-control, security and verification approaches.

It aims to identify:

- What each approach is documented to do.

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- Where its intervention occurs.
- What type of assurance it provides.
- What evidence is available.
- Which questions remain unresolved.

### 3.2 Scope Limitations

This is a representative landscape review, not a complete inventory of all AI governance products, research projects, standards or regulations.

The inclusion of a system does not imply endorsement.

The absence of a documented feature in the reviewed sources does not prove that the feature is absent from the system.

Product documentation, research publications, open-source software, standards and regulations are treated as different types of evidence.

The report does not perform controlled product testing, source-code security audits, independent benchmarking or legal compliance assessments.

## 4 Methodology and Evidence Classification

### 4.1 Evidence Approach

The comparison prioritizes primary documentation, including:

- Official technical documentation.
- Official research publications and model reports.
- Standards and regulatory texts.
- Official project repositories and technical descriptions.
- OIP source documents and recorded research limitations.

Secondary commentary is not treated as equivalent to independent technical verification.

### 4.2 Evidence Labels

| Label | Meaning |
|---|---|
| Supported | The reviewed source directly supports the stated point. |
| Partially Supported | Some relevant evidence exists, but it does not establish the complete claim. |
| Not Established | The available evidence does not establish the claim. |
| Not Verified | The relevant source or technical behaviour has not been independently verified for this review. |
| Not Applicable | The dimension does not directly apply to the approach. |

### 4.3 Comparison Discipline

A documented feature is not automatically proof of effectiveness.

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A vendor statement is not an independent performance result. A research prototype is not automatically a production system. A passing software test is not scientific validation.

These distinctions apply equally to OIP and the other approaches reviewed here.

## 5 Landscape of Selected Approaches

### 5.1 AI Guardrails and Model Safeguards

#### NVIDIA NeMo Guardrails

NeMo Guardrails provides configurable rails for controlling large language model behaviour. Its documented architecture includes input, retrieval, dialog, execution and output control areas.

It should not be characterized solely as an output filter. The exact behaviour depends on the configured rails and implementation [1].

#### Meta Llama Guard

Llama Guard is a family of safety models intended to classify potentially unsafe inputs and outputs according to defined safety categories.

Its classification role should not be confused with a complete runtime authorization system for arbitrary external actions [2].

#### Microsoft Azure AI Content Safety

Azure AI Content Safety provides content-related safety capabilities. Groundedness detection evaluates whether model responses are supported by supplied source material.

The cited documentation states that this groundedness detection feature is scheduled for retirement on 30 April 2027. This date applies to that feature, not necessarily to every Azure AI Content Safety capability [3].

Groundedness evaluation should not be treated as proof of general factual correctness or complete agent authorization.

### 5.2 Runtime and Agent Security

#### Microsoft Agent Governance Toolkit

The Microsoft Agent Governance Toolkit describes governance capabilities for agentic systems. Its documented features and maturity should be assessed against the specific release and deployment context [4].

#### NVIDIA OpenShell

NVIDIA OpenShell documents sandboxed execution environments for autonomous AI agents, with declarative policies and permission controls.

Execution isolation addresses a different security boundary from content moderation. The actual assurance depends on the configured environment and deployment controls [5].

#### CrowdStrike Falcon Guardian

CrowdStrike announced an AI Kill Switch in Falcon Guardian on 3 November 2026. The announcement describes agent-related security intervention capabilities.

This is a vendor-described capability. It should not be interpreted as independent proof that every agent, environment or possible action can be controlled [6].

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## 6 Formal Verification and Technical Assurance

Formal verification attempts to establish whether a system satisfies defined mathematical properties under specified assumptions.

It is not equivalent to general-purpose AI governance.

### 6.1 Marabou

Marabou is a tool for reasoning about neural networks and specified properties. Its results are bounded by the supported models, query formulation and assumptions [7].

### 6.2 alpha-beta-CROWN

alpha-beta-CROWN is a neural-network verification approach for checking specified properties under supported verification conditions. It is not a universal verifier for all AI agents, tools or deployment environments [8].

### 6.3 Assurance Boundary

A verification result is meaningful only in relation to:

- The model or system being verified.
- The property being tested.
- The input domain and assumptions.
- The verification method and its limitations.
- The validity of the implementation being assessed.

A proof about a restricted model property is not a universal proof about an AI system.

## 7 Governance Standards and Regulation

### 7.1 NIST AI Risk Management Framework

The NIST AI Risk Management Framework provides a voluntary framework for identifying, assessing and managing AI-related risks.

It is a governance and risk-management framework, not a specific runtime agent-interception product [9].

### 7.2 ISO/IEC 42001

ISO/IEC 42001 specifies requirements for establishing, implementing, maintaining and continually improving an AI management system within an organization.

It concerns organizational management processes and should not be confused with proof that every AI output or agent action is safe [10].

### 7.3 EU AI Act

The EU AI Act establishes legal requirements for AI systems within its regulatory scope.

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Article 14 addresses human oversight for high-risk AI systems. It should not be described as a universal technical kill switch or as a requirement that every AI system use one particular runtime architecture [11].

### 7.4 Different Types of Assurance

- Governance frameworks guide risk-management practices.
- Management-system standards define organizational requirements.
- Regulations create legal obligations within their scope.
- Runtime controls constrain system behaviour during execution.
- Formal verification establishes specified properties under stated assumptions.

These approaches may complement one another, but they are not substitutes.

## 8 Comparative Analysis

The following matrix summarizes the principal intervention areas of the selected approaches.

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| Approach | Primary Focus | Intervention Area | Assurance Boundary |
|---|---|---|---|
| NeMo Guardrails | Conversational AI controls | Input, retrieval, dialog, execution and output rails | Configuration and implementation dependent |
| Llama Guard | Safety classification | Input and output classification | Not complete action authorization |
| Azure AI Content Safety | Content safety and groundedness | Content and response assessment | Does not establish general factual correctness |
| Microsoft Agent Governance Toolkit | Agent governance and controls | Agent policy and governance workflows | Release and deployment dependent |
| NVIDIA OpenShell | Execution isolation | Agent execution environment | Isolation does not establish model correctness |
| CrowdStrike Falcon Guardian | Agent-related security | Product-specific runtime intervention | Vendor-described capability; scope must be checked |
| Marabou | Neural-network verification | Specified network properties | Bounded by supported models and assumptions |
| alpha-beta-CROWN | Neural-network verification | Specified mathematical properties | Not a universal agent verifier |
| NIST AI RMF | Risk management | Organizational risk processes | Framework, not runtime enforcement |
| ISO/IEC 42001 | AI management system | Organizational management processes | Not proof of individual output safety |
| EU AI Act | Legal governance | Requirements within regulatory scope | Legal obligations, not one universal architecture |
| OIP | Evidence qualification methodology | Evidence, measurement and authorization stages | Research-stage; validation not established |

*This matrix is a high-level classification. It is not a benchmark, performance ranking or independent certification of the listed systems.*

## 9 Evidence, Missingness and Decision Authorization

A critical issue in evidence-based decision systems is the treatment of incomplete or unsupported information.

A missing value does not necessarily represent zero. An unavailable observation does not establish a negative outcome. An unverified claim should not automatically be treated as an approved fact.

A robust evaluation process should distinguish:

- Observed evidence.
- Missing evidence.
- Invalid or inconsistent evidence.

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- Evidence that has not been independently verified.
- Evidence that is insufficient for the intended decision.

### 9.1 Decision Authorization

A useful research question is whether a system should be permitted to make a decision when required evidence is incomplete.

A conservative decision process can distinguish:

1. Authorization supported by sufficient evidence.
2. Pending review because evidence is incomplete.
3. Blocked because a required condition is not satisfied.
4. Invalid input or inconsistent evidence.

This is a proposed analytical structure, not evidence that all reviewed systems lack such controls.

### 9.2 Economics of Abstention

Refusing or deferring a decision can create costs, including review time, delay and operational overhead.

Approving an unsupported decision can also create costs, including incorrect action, downstream damage and remediation.

A future empirical evaluation could compare:

- Cost of false approval.
- Cost of false refusal.
- Cost of human review.
- Cost of decision delay.
- Cost of implementation and maintenance.

No quantified economic advantage for OIP is established by this report.

The relationship between abstention, selective prediction, risk and decision cost is an existing research area. OIP should not claim ownership of the general concept.

## 10 OIP: Architecture and Research Position

### 10.1 Research Objective

The Organic Intelligence Protocol (OIP) is a research-stage methodology concerned with evidence qualification, measurement authorization, missingness control and fail-closed decision rules.

Its stated objective is to prevent unsupported evidence or incomplete measurements from being silently converted into authorized conclusions.

### 10.2 Conceptual Architecture

The documented OIP architecture has been described through three broad components:

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- Human-Derived Intelligence (HDAI) layer.
- Cognitive Governance Workflow (CGW) layer.
- Arbitration layer.

Associated research concepts include:

- Grounded Feedback Loop (GFL).
- Infrastructure Independence (II).
- Adaptive Moral Logic (AML).

These concepts should not be treated as fully operationalized or empirically validated merely because they appear in the architecture.

### 10.3 Mathematical Reference Formulations

The OIP research record contains the following reference formulation for Score:

$$Score = 0.6(OE) + 0.3(CV) + 0.1(VIS)$$

However, a historical additive-versus-multiplicative formulation conflict remains unresolved in the source record.

For that reason, this report does not designate the displayed additive expression as an independently confirmed canonical equation.

The documented SIS reference formulation is:

$$SIS = 0.5(RC) + 0.3(AFL) + 0.2(CAS)$$

Infrastructure Independence is expressed as:

$$II = 1 - IDS$$

A documented stability expression is:

$$Stability = GFL \times (1 - IDS) \times AML$$

A separate expression is:

$$LIV = \frac{RI \times SR \times AR}{IDS}$$

For this expression, the recorded boundary condition is:

$$IDS = 0 \Rightarrow LIV = UNDEFINED$$

No epsilon substitution or denominator replacement is authorized by this boundary rule.

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These equations are reference formulations. Their inclusion does not establish that all variables have completed measurement protocols or that all weights are empirically calibrated.

### 10.4 Mathematical Boundary Limitation

The boundary condition $IDS = 0$ makes the displayed LIV expression undefined.

It does not, by itself:

- Establish data integrity.
- Prove that infrastructure dependence is absent.
- Establish that all other OIP variables equal one.
- Demonstrate that neural-network gradients stop.
- Prove scientific validity or operational safety.

Those are separate claims requiring separate definitions, tests and evidence.

### 10.5 Current Research and Validation Status

The OIP v1.0.40 research-stage record identifies unresolved methodological and evidence issues [12].

| Area | Recorded Status |
|---|---|
| Candidate engine | Candidate, not approved |
| Measurement authorization | Not granted |
| Scientific validation | Not established |
| Production authorization | Not authorized |
| Commercial readiness | Not assessed |
| Mathematical specification | Open issues remain |
| Independent validation | Not established by this report |

Reported software test passes are evidence about the tested software behaviours and test cases. They do not establish scientific validity, real-world safety or general performance.

### 10.6 Implementation Boundary

OIP v0.2.0 is a limited synthetic Python reference implementation demonstrating selected authorization rules. It should not be represented as a complete implementation of the broader OIP v1.0.x methodology [13].

The presence of a reference implementation does not establish that every conceptual component has been implemented or empirically tested.

### 10.7 Research Boundary

OIP should not currently be represented as:

- A scientifically validated universal AI safety system.
- A fully implemented runtime agent interception layer.
- A proven replacement for existing guardrails or security tools.

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- A system with established commercial performance.
- A demonstrated solution to all forms of AI hallucination.

Its present status is research-stage and evidence-bounded.

## 11 Open Research Questions

The comparison identifies several questions that require further technical and empirical work.

1. How should evidence sufficiency be defined for different classes of AI decisions?
2. How can missingness and invalid evidence be handled consistently across heterogeneous datasets?
3. Which authorization rules can be formally specified and independently tested?
4. How should runtime permissions interact with model-level guardrails and external security controls?
5. What is the measurable cost of abstention compared with unsupported approval?
6. How can independent evaluators reproduce reported results from frozen source material?
7. Which OIP variables have sufficiently complete operational definitions for measurement?
8. Which mathematical formulations require formal owner approval before implementation or evaluation?

These questions define a research agenda. They are not claims of demonstrated OIP capability.

## 12 Recommended Research and Validation Roadmap

A defensible development sequence is:

1. Complete the source audit and mathematical completion register.
2. Resolve historical formula conflicts and obtain explicit owner decisions.
3. Define operational measurement protocols for unresolved constructs.
4. Freeze the specification, codebook, boundary rules and missingness policy.
5. Obtain independent review using the same frozen source pack.
6. Implement executable mathematical and software tests for the approved specification.
7. Record observed outputs, failures and unresolved issues without silently changing test criteria.
8. Conduct appropriately designed empirical studies before making scientific or performance claims.
9. Maintain version control, provenance records, change history and reproducibility documentation.

Completion of a software milestone should not automatically advance the scientific validation status.

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## 13 Conclusions

The reviewed landscape contains multiple approaches to AI governance and control, but they address different technical and organizational problems.

Content safeguards, runtime execution controls, formal verification, risk-management frameworks and regulatory requirements should be evaluated according to their actual scope.

No single comparison dimension is sufficient to establish overall system safety.

OIP contributes a research-stage methodology focused on evidence qualification, measurement authorization, missingness control and fail-closed decision discipline.

Its unresolved measurement protocols, mathematical specification questions and lack of established scientific validation remain material limitations.

The appropriate next step is not to claim superiority or universal applicability. It is to complete the specification, make its constructs measurable, test the implementation and obtain independent empirical evidence.

The conclusions of this report are limited to the reviewed documentation and the stated evidence boundaries.

## Final Evidence and Authorship Statement

This report was authored by Sudharsan Das, Lead Architect of the Organic Intelligence Protocol.

The report documents a comparative research review and the current research-stage position of OIP.

Authorship identifies responsibility for this document and the OIP research documentation described herein. It does not constitute independent scientific validation, formal certification, regulatory approval or third-party endorsement.

Descriptions of external systems are limited to the identified sources. Product capabilities may change by version, configuration and deployment environment.

OIP claims are limited to the documented research record. Unresolved claims remain unresolved until supported by reproducible evidence.

This report is not a certification, independent security audit, formal verification certificate, regulatory opinion, scientific validation report or product endorsement.

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Where a source does not establish a claim, the claim should be narrowed, labelled as unverified or removed.

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**Sudharsan Das**

Lead Architect, Organic Intelligence Protocol

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